Department of Electrical Engineering
University of Arkansas

ELEG 46003/51703L Digital Signal Processing

Ch. 0 Preface

Dr. Jingxian Wu

wuj@uark.edu

GENERAL INFORMATION

- **Instructor: Dr. Jingxian Wu**
  - Email: wuj@uark.edu
  - Phone: (479) 575-6584
  - Office hour: Tue 12:30-1:30

- **Lecture Schedule**
  - Tue. Thu. 11 – 12:15

- **Lab Schedule**
  - Thu. 11 – 12:15 (approximately once every two weeks)

TEXTBOOK AND REFERENCES

- **Textbook (Optional)**
  - Luis F. Chaparro, *Signals and Systems Using Matlab*, Academic Press, 2010.

- **Required Software**
  - Matlab

- **References (optional)**
  - S.S. Soliman, M.D. Srinath, *Continuous and discrete signals and systems*, 2nd Ed., Prentice Hall, 1998.

COURSE INFORMATION

- **Pre-requisite:**
  - Signals and Systems

- **This course involves heavy mathematical derivations**
  - A large number of examples will be given in class – It's ESSENTIAL for you to repeat all the examples by yourself after class.

- **Teaching format**
  - Slides
  - Examples
  - Exercises
  - Homework assignments (problems, software)
  - Labs

TENTATIVE SCHEDULE

- Week 1: Ch.1 Discrete-Time Signals and Systems
- Week 2: Ch. 1 Discrete-time Signals and Systems
- Week 3: Ch. 1 Discrete-Time Signals and Systems
- Week 4: Ch. 2 The Z-Transform
- Week 5: Ch. 2 The Z-Transform
- Week 6: Ch. 2 The Z-Transform
- Week 7: Ch. 2 The Z-Transform (**Test 1 on 3/5 Thu.**)
- Week 8: Ch. 3 Discrete-Time Fourier Transform
- Week 9: Ch. 3 Discrete-Time Fourier Transform
- Week 10: Ch. 3 Discrete-Time Fourier Transform
- Week 11: Ch. 3 Discrete-Time Fourier Transform
- Week 12: Ch. 4 Discrete Fourier Transform
- Week 13: Ch. 4 Discrete Fourier Transform (**Test 2 on 4/9 Thu.**)
- Week 14: Ch. 4 Discrete Fourier Transform
- Week 15: Ch. 5 Discrete-Time Filters
- Week 16: Ch. 5 Discrete-Time Filters
- Week 17: **Test 3 in the week of 5/4**

GRADING POLICY

- **Grades Percentage**
  - Test 1 — 24%
  - Test 2 — 24%
  - Test 3 — 24%
  - Homework — 14%
  - Lab — 14%

- **Grades**
  - A: $90 \leq \text{grade} \leq 100$
  - B: $80 \leq \text{grade} < 90$
  - C: $70 \leq \text{grade} < 80$
  - D: $60 \leq \text{grade} < 70$
  - F: $0 \leq \text{grade} < 60$

GRADING POLICY

- **Due dates for homework and lab report will be strictly enforced. Late submission within one week after due will receive a 20% deduction; no credit if submitted one week past due.**

- **If for some legitimate reason (sickness, death in the family, etc.), you cannot take an exam on the scheduled day, you must notify the instructor prior to the exam.**

ONLINE RESOURCES

- **Course Home Page**
  - https://wuj.hosted.uark.edu/teaching/eleg5173/eleg5173.html
  - All the course related materials, such as slides, homework assignments, links, announcements, labs, etc., will be posted on this website.
  - Please check the webpage regularly (at least once per week) for update.

ADDITIONAL ISSUES

- **Academic Honesty**
  - Each University of Arkansas student is required to be familiar with and abide by the University's 'Academic Integrity Policy' which may be found at http://provost.uark.edu/
  - Any kind of activities related to academic dishonesty (copying homework, lab report, code, plagiarism, etc.) will be dealt with.
  - If you are not sure about plagiarism, please contact the instructor.
- **Questions are welcome in my class**
  - You are very welcome to raise any question related to course materials.
  - Please feel free to stop me at any time if you have any question.
  - You can also ask me question via email or during office hours.
- **To respect your fellow students as well as the instructor, please turn off or silencing your cell phone.**
  - No text messaging or web surfing!
- **BE ON TIME!**
- **Have Fun!**